

Literature Review and Reference Management: Predictive Cold Chain Disruption Modeling for Pharmaceutical Logistics

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Abstract

Lab 2: Literature Survey Report

Key References & Resources:

Base Paper: Xie et al. (2025)

Dataset: Cold Chain Shipment Silent Failure Dataset (Kaggle)

1. Introduction to the Research Problem

Cold-Chain Logistics for Pharmaceuticals is an extremely sensitive area where short periods of time outside of temperature-controlled conditions can make life saving drugs/vaccines/biologic useless or harmful. An estimated 50% of all vaccines are lost (wasted) worldwide on a yearly basis; a substantial number of these lost vaccines are due to temperature control issues that occur when transporting & storing vaccines. Temperature related loss has been estimated by pharmaceutical companies to be a staggering \$35+ Billion dollars each year in terms of revenue. Temperature controlled product failures do not just have an economic impact. The implication if some compromise pharmaceutical products make it into patient's hands is the occurrence of adverse events, decreased patient confidence in the healthcare system and potential for regulatory fallout for manufactures and distributors.

The traditional approach for monitoring in a cold chain is based upon using static thresholds with an alarm generated by alerting systems once a sensor measurement surpasses a pre-determined value (for example when the temperature exceeds 8°C). Although these types of systems are relatively easy to create; however, there exist two significant problems. Firstly, the excessive number of false positive alarms caused by transient sensor noise, opening doors while the products were being loaded into the container, and/or environmental fluctuations will cause "alert fatigue" for logistics operations personnel. Secondly, and perhaps more damagingly, these threshold-based systems do not have the ability to identify what is called "silent failures", which can occur as a result of long-term thermal damage to the product quality, the existence of temperature differences in space throughout the container(s), and/or bias in how sensors are placed relative to the location of potential temperature deviations. Silent failures are considered the most hazardous type of failure of a cold chain because they go completely undetected.

Internet of Things (IoT) sensors have rapidly expanded the number and granularity of cold-chain data available for post-

shipment analysis. Contemporary shipments could now potentially have data streams for multiple dimensions including temperature, humidity, light exposure, door events, vibration and gps coordinates collected every thirty seconds! With such a density of information, modern ML/DL can advance past simple threshold breaches of shipment parameters to actually predictive failure analysis of impending disruptions. But introducing 'black box' ML for pharma can have significant drawbacks related to trust, compliance and accountability. What action needs to be taken with a shipment identified as high risk? Data managers and QA teams need to understand why it was identified, making Explainable AI (XAI) approaches like LIME and SHAP a priority in ML-based cold-chain systems.

The literature survey presented here discusses 20 papers published between years 2021 - 2025, and all of these articles can be classified and categorised within four main themes representing the proposed research topic: cold-chain anomaly detection, IoT-based monitoring systems, pharmaceutical supply chain intelligence, machine learning-based time-series predictions, and explainable AI for logistics decision support. The research contribution can thus be contextualised with the work by Xie et al. [1] as a reference paper. We want to develop on the work in the base paper to provide a new research contribution of developing a machine learning-based predictive explainable model for Cold-Chain Disruptions modeling.

2. Literature Review

The 20 surveyed papers naturally form 5 subject matter categories namely (a) cold-chain anomaly detection and monitoring, (b) Isolation Forest and modern anomaly detection methods, (c) cold-chain time series and deep learning, (d) pharmaceuticals and vaccine cold-chains, and (e) explainable AI, predictive risk, and support to decision making. Section subsections briefly analyze the contained papers from each category.

2.1. Cold-Chain Anomaly Detection and Monitoring

The work done by Xie et al. [1] [BASE PAPER] focuses exactly on the main problem - detection of anomalies in CCL data streams. The authors analyzed CCL data streams, defined data flow using sliding window and proposed mathematical models for flow control and Correlation Coefficients for detecting three type of abnormal events. This method contributes an effective enhanced iForest (eiForest) that uses subsampling and a cross factor for overcoming the swamping and masking issues faced by the classical iForest. Experimental study and evaluation of the proposed scheme results in performance such as Average F1-score of 0.8639 and average AUC of 0.9064; where this method gives the performance even higher than SVM, LOF and the classical iForest; specifically on large dimensionality of data but the processing time was higher than classical iForest and no existing open source data sets are taken for performance validation.

The Gillespie et al. [2] study develops a working prototype for real-time cold chain logistics anomaly detection using an IoT-based system. As a bonus for practitioners, their research utilizes commercial, real-world transportation trucks moving food products within Northern Ireland and Ireland, deploying Digital Matter Eagle cellular data loggers. They provide a real-world proof of concept that integrating trip and movement detection with environmental thresholds significantly minimizes false positive alerts compared to systems employing only static thresholds. Their research addresses practical implementation problems with sensor calibration, although it does not utilize machine learning classifiers like k-NN or offer advanced forecasting capabilities.

Addressing the false temperature alarm problem, Meng et al. [3] identify another interesting challenge in cold chain operations. They propose a cost-effective over-temperature alarm system specifically designed for cold chain delivery. By prioritizing real-time alerts without relying on complex neural networks, the system effectively mitigates food loss while remaining economically viable. While this method focuses heavily on cost-effective alerting, its scope is narrower compared to full anomaly detection frameworks.

Following our discussion on monitoring, Cil et al. [4] introduce IoT into the container-port scenario to showcase real time IoT based monitoring resulting in a 100% container viewability with 2-min container excursion detection, and 82% reduction of manual inspection time. Although based on a threshold detection method without IoT for anomaly inspection based on machine learning techniques, the IoT device in Cil et al. [4] provides an empirical confirmation for the practical value of IoT infrastructure in a cold-chain setting and suggests that it should also involve IoT for deeper analysis using machine learning techniques beyond thresholds.

2.2. Isolation Forest and Advanced Anomaly Detection Algorithms

A natural follow-up with a fundamental novelty. Xu et al. [?] developed Deep Isolation Forest (DIF) by integrating representation learning by neural networks and random-partition

based isolation. DIF maps the data into an optimized representational space where it is easier to isolate anomalies. They achieved an AUC score average of 0.943 on 10 benchmark datasets, which are higher than iForest, RRCF, COPOD, and DeepSVDD, DAGMM. The new method is highly related because the proposed better version of iForest could adopt DIF for learning representations, and it's especially crucial when dealing with non-linear relationships often observed in multi-sensor cold-chain data. Nonetheless, DIF costs much more computation time to deploy in real-time applications.

The Kaggle Cold Chain Shipment Silent Failure Dataset will be the critical data asset for this area of study. This dataset comprises 8,000 shipment entries containing data on temperature, humidity, vibration, door status, and failure outcomes. It is specifically designed as a benchmarking dataset for models attempting to detect silent failures—instances where shipment quality declines without triggering a conventional alert. The dataset provides an objective benchmark to evaluate predictive models. This dataset serves as the defined validation benchmark for the current project, with the baseline figures used to set the performance standards the proposed explainable predictive models will seek to surpass.

2.3. Deep Learning for Cold-Chain Time-Series

Chen et al. [5] have managed to deploy LSTM-Autoencoder models to embedded devices with limited computational capabilities (ESP32) to be able to detect anomalies in vaccine refrigerators in real-time. This TinyML application presents a good F1-Score of 0.94 with latency < 100 ms and a model footprint of 48 KB, showing how deep learning on the edge can be used for cold-chain applications. This is related to the project presented here, since the LSTM model deployed on the devices is able to detect anomalies at the source where the data is being generated, avoiding communication to the cloud.

In Zhang et al. [6], authors also leverage self-attention based Transformer models and apply on cold-chain logistics time series data. They design an effective model based on Transformer that addresses challenges in LSTM for long-term sequential dependencies, and achieved an F1-Score of 0.92, Precision of 0.90, Recall of 0.94, and AUC of 0.96. The Transformer model yields 4% F1-Score gain against the LSTM-Autoencoder. The inherent temporal explanation of Transformer by highlighting which time series elements are responsible for an anomaly prediction seems implicitly introduced although SHAP for the Transformer has not yet been introduced.

For example, Luo et al. [7] developed a K-Medoids-LSTM-XGBoost based ensemble model. This multi-model strategy uses spatial clustering (K-Medoids) for temperature segmentation and combines temporal predicting (LSTM) with residual correction (XGBoost) to predict the cold chain loading environment for agricultural products. The model achieved a Mean Absolute Error (MAE) of 2.5343, demonstrating that combining different models from across families is a highly effective approach to capturing complex dependencies.

To tackle these challenges with realistic cold-chain sensor data, Martinez et al. [8] propose GAN-based denoising to achieve improvements in signal to noise ratio up to 12 dB, and

an anomaly detection F1-Score of 0.91 and 15-min predictive alert time. A common practical limitation on academic solutions is sensor noise encountered in actual cold-chain settings that often degrades anomaly detection performance. The GAN based pre-processing could be adapted in the proposed research pipeline as a data quality enhancement procedure.

2.4. Pharmaceutical and Vaccine Cold-Chain Intelligence

In their paper “An IoT-enabled system for managing the vaccine supply chain sustainable in resource-limited developing country”, Jiang et al. [9] propose an IoT system for sustainable vaccine cold-chain management and report a temperature compliance rate of 98.5% with a 73% decrease in manual data logging in real-world settings. Their study supports the business case of leveraging IoT for logistics and supply chain operations, however it does not take full advantage of IoT by providing the ML-based prediction of disruptions. In fact, they themselves acknowledge the integration of ML-driven predictive insights as a crucial area of future research and a motivation for future work, hence direct relevance for the work proposed herein.

Similarly to Jiang et al., Hu et al. [10] introduce an intelligent vaccine supply chain management system that includes blockchain (Hyperledger Fabric), IoT for temperature monitoring, and a Random Forest algorithm for demand forecasting. Their results include 8.2% MAPE for demand forecasts and 450 TPS for blockchain transactions with 99.7% integrity of data. However, despite using blockchain to guarantee the integrity of pharmaceutical supply chain data, anomaly detection is based on a relatively basic rule-based system, and the model has limitations concerning individual prediction explainability. Finally, Hezam et al. [11] explore the applicability of AI techniques in drug delivery within healthcare supply chains. Their machine learning models are used for delivery time prediction, achieving 93.7% prediction accuracy, and the study shows that temperature is the second most important feature for delivery prediction.

Therefore, their study corroborates the focus on temperature as key predictor that should be exploited for temperature-aware logistics predictive modeling. Yet, they do not consider real-time monitoring data from an IoT infrastructure, and neither the system they propose includes ML-driven explanations for individual predictions.

2.5. Explainable AI, Predictive Risk, and Decision Support

Islam et al. [12] proposes an explainable ML framework with SHAP and LIME for delivery delays in smart supply chains. Using XGBoost, they achieve a 96.3% accuracy with a 0.95 F1-Score. SHAP explains their predictions, identifying weather severity, distance, and shipment weight as the three top risk factors. The proposed framework is most similar in nature to the explainability goals of the present research, demonstrating the potential of SHAP for understanding complex supply chain risk models at a feature-level.

However, the framework does not utilize specific data characteristic of cold-chain operations and only addresses general supply chain data. Li et al. [13] employs SHAP-based explainability within the domain of predictive maintenance in industrial

IoT. Their LightGBM model employing SHAP TreeExplainer can successfully predict remaining useful life with an R value of 0.94 and can attribute sensor reading deviations to specific degradation features. Methodologically, this work offers valuable guidance in the application of SHAP to cold-chain components, such as refrigerators and other cooling devices. In the future, a similar approach will be adapted for explaining sensor-based predictions of failure and deviations of cold chain equipment.

Konovalev and Ludwig [14] compare eight ML classifiers on a dataset for temperature alarm classification within pharmaceutical supply chains. XGBoost proves to be the most adept classifier in this task, achieving 97.8% accuracy and an F1-Score between 0.92 and 0.97. The analysis of this research provides a useful benchmark against which future algorithm performance can be compared, and validates the effectiveness of gradient boosting algorithms in predicting temperature deviations within a logistics setting, such as that of a pharmaceutical supply chain.

Sattar et al. [15] provides a broader comparative study that assesses eight ML algorithms with a cost-accuracy trade-off for predictions on the DataCo Smart Supply Chain Dataset. LightGBM, with 97.1% accuracy, is shown to train in a considerably shorter time (3.2 seconds versus 47 seconds for neural networks), demonstrating the practicality of tree-based ensembles for such applications. This supports the focus on gradient boosting methods for the proposed cold-chain disruption model.

Wu et al. [16] proposes a digital-twin service platform for cold-chain logistics and achieve a temperature prediction RMSE of 0.42°C and a 30% reduction in response times to excursions by leveraging an IoE connectivity and real-time simulation. To supplement this, Nozari et al. [17] introduces a system that, using AIoT, minimizes cold chain operating costs by 26% and transportation expenses by 60%. Douaioui et al. [18] presents a systematic survey and categorization of ML and DL techniques for demand forecasting applications in supply chains, identifying that the typical error reduction (MAPE) for LSTM and CNN-LSTM are between 15% and 25% compared with classical approaches, and is regarded as the state-of-the-art in the field. This work is helpful as it offers insights into the implementation of DL models for this area.

3. Comparative Analysis of Existing Methods

Here is a side-by-side analysis of the 20 discussed papers across the five critical categories (i.e. Core ML/DL method, reactive vs predictive type detection strategy, explainability in use, dataset classification, best presented performance measure), presented on Table 1. While there’s evidence of a movement from threshold-based reactive monitoring towards ML for a predictive approach, the research still lacks on integrating explainability with predictive cold chain models.

Key observations from the comparative analysis:

- **Reactive vs. Predictive:** 10 papers attempt anomaly detection after the anomaly event has occurred (reactive anomaly

Table 1: Comparative Analysis of 20 Reviewed Papers

#	Study	Primary Method	Approach	XAI	Best Metric
1	Xie et al. (2025) ★	Improved iForest	Reactive	No	F1: 0.86, AUC: 0.91
2	Gillespie et al. (2023)	IoT + Trip Detect	Reactive	No	False Alerts ↓ 100%
3	Wu et al. (2023)	LSTM + Digital Twin	Predictive	No	RMSE: 0.42°C
4	Meng et al. (2024)	IoT Alarm System	Reactive	No	Cost ↓ 58%
5	Xu et al. (2022)	Deep Isolation Forest	Reactive	No	AUC: 0.943
6	Jiang et al. (2024)	IoT framework	Reactive	No	Compliance: 98.5%
7	Hu et al. (2023)	RF + Blockchain	Predictive	No	MAPE: 8.2%
8	Kaggle Dataset (2024)	Sensor Data	Reactive	No	8,000 Records
9	Nozari et al. (2025)	Meta-heuristic opt.	Predictive	No	Cost ↓ 26%
10	Chen et al. (2024)	LSTM-Autoencoder	Reactive	No	F1: 0.94
11	Konovalenko (2021)	Gradient Boosting	Reactive	No	Acc: 95.9%
12	Zhang et al. (2024)	Transformer	Reactive	Implicit	F1: 0.92, AUC: 0.96
13	Islam et al. (2025)	XGBoost + SHAP/LIME	Predictive	Yes ✓	Acc: 96.3%
14	Li et al. (2023)	LightGBM + SHAP	Predictive	Yes ✓	R ² : 0.94
15	Sattar et al. (2025)	LightGBM + 7 MLs	Predictive	No	Acc: 97.1%
16	Douaioui et al. (2024)	Review (LSTM, CNN)	N/A (Review)	No	MAPE ↓ 15%
17	Luo et al. (2024)	K-Med-LSTM-XGBoost	Predictive	No	MAE: 2.53
18	Liang et al. (2025)	GAN + Signal Proc.	Predictive	No	SNR ↑ 12 dB
19	Hezam et al. (2024)	RF + GB + NN	Predictive	Partial	Acc: 93.7%
20	Cil et al. (2022)	IoT + Threshold	Reactive	No	Visibility: 100%

detection), with only 8 attempting anything approaching prediction or a forecasting methodology (predictive analytics). Only two papers attempt to bridge the two concepts, with Islam et al. [12] and Li et al. [13] both incorporating an explanation element with predictive techniques.

- **Explainability Gap:** Only two papers (Islam et al. [12], Li et al. [13]) use explainability techniques (SHAP, LIME). Two papers (Zhang et al. [6], Hezam et al. [11]) rely on Transformer’s implicit attention and partial feature importance analysis, respectively. For the other sixteen papers, machine learning models act like black boxes.
- **Cold-Chain Specificity:** Papers 1–4, 8, 10, 17, 18, 20 examine exactly cold-chain supply chains. Paper 13–14 examines generic supply chains and applies XAI, but without mention of cold chains. We see a definite deficiency in the intersection of prediction within cold chains and the explainability of those predictions. Dataset Availability: Only Skarin et al. [19] use a publicly available cold-chain dataset with failure labels. Most other papers use proprietary or simulated data, limiting reproducibility.
- **Performance Ceiling:** Models that work the best reach an F1-Score of 0.89 to 0.94 for detection of anomalies in cold supply chain, and between 0.95 to 0.97 in case of predicting overall supply chain outcome. In our opinion this also leaves considerable scope of improvement via sophisticated feature engineering and models design.

4. Identified Research Gaps

Based on the comprehensive analysis of 20 papers, the following research gaps have been identified that collectively motivate the proposed research topic:

- **Gap 1: Reactive vs. Predictive Detection:** However, most cold-chain anomaly studies in existing works (Xie et al. [1], Gillespie et al. [2], Meng et al. [3], Cil et al. [4]) aim for the detected anomaly only after its occurrence. It’s relatively rare to predict cold-chain disruption prior to any damage on the quality of cold-chain products. Change from anomaly detection towards prediction-based disruption modeling is crucial in providing proactive measures, such as re-routing the products, changing refrigerator temperature, prioritizing the inspections.
- **Gap 2: Explainability Deficit in Cold-Chain ML:** Although SHAP and LIME have been used to supply chain prediction problems in general (e.g., Islam et al. [12]; Li et al. [13]), none of these have explicitly investigated XAI techniques for cold-chain disruption prediction. Logistics operators and quality assurance departments require insights from the predicted disruption in terms of sensor patterns, characteristics of the shipment, or the external conditions that influence disruption predictions so that the correct measures can be taken.

- **Gap 3: Silent Failure Detection:** Silent failures that are undetectable by any sensors - while slowly compromising product quality - are the most detrimental class of cold-chain failure. Unfortunately, while the Kaggle Silent Failure Dataset exists (Skarin et al. [19]), existing academic literature doesn't extend advanced machine learning approaches and explainability techniques to this particular problem. The provided Kaggle baseline (GB F1: 0.89) suggests appropriate objectives that more advanced models should try to improve on.
- **Gap 4: Multi-Sensor Feature Engineering:** Currently most cold-chain analysis projects utilize temperature as the sole feature. Given the richness of multi-modal sensor information captured from up-to-date IoT devices such as humidity sensors, door open/closed signals, accelerometers and GPS sensors, is greatly untapped. The ability to extract such information through feature engineering relating to trends, spatial relations, and interaction between different sensor channels would likely further boost the prediction capabilities.
- **Gap 5: Isolation Forest Enhancement with Explainability:** An enhanced iForest in the base paper reaches good performance (F1: 0.86, AUC: 0.91), yet the obtained anomaly score is hard to explain. Deep Isolation Forest, from Xu et al. [?], gets higher detectability but no further explainability is provided. The integration of the enhanced isolation method with SHAP based feature explanation for the model could make it an interpretable anomaly detection framework - a method that has not been tested yet.
- **Gap 6: Benchmark Standardization:** Unfortunately, there is currently a lack of uniform, publicly accessible cold-chain datasets containing information on possible failure modes for any reasonable comparison. Almost all existing papers provide results from proprietary or simulated cold-chain datasets, so no two methods can directly be compared for example.
- **Gap 7: Pharmaceutical-Specific Constraints:** Little research discusses explicitly the unique requirements to pharmaceutical regulations (e.g. GDP, FDA 21 CFR Part 211) requiring not only a detected but also recorded and explained temperature excursion. By leveraging on XAI, the excursion reports with a causal explanation may be automatically drafted in compliance to the regulation.
- **Gap 8: End-to-End Pipeline Integration:** Current research largely covers these individual pieces (sensor hardware, anomaly detection algorithm, and blockchain for trace) as separate concerns. Instead we focus on creating an integrated pipeline for IoT data ingestion, feature engineering, predictive modeling, explainability, alerting and more.

5. Motivation for Selecting the Base Paper

The base paper by Xie et al. [1], 'An Anomaly Detection Scheme for Data Stream in Cold Chain Logistics,' was selected

as the basis for this study due to a variety of compelling factors. First, it directly tackles the primary technical problem – the anomaly detection of data stream in cold chain logistics – using a well-established and easily-adaptable algorithm (Isolation Forest). The sub-sampling and cross factor improvements made to the base iForest demonstrated that the development of algorithms could improve the detection efficiency. (F1-score 0.8639, AUC 0.9064) is used as a basis for future study.

Secondly, the framework presented in the base paper can be applied to this proposed study in terms of its data processing, such as its Sliding Window Correlation and data stream architecture. Data stream and its infrastructure from iForest could accept the multi-sensor attributes from the Kaggle dataset (such as temperature, humidity, vibration and door open/close Status) as input. This algorithm can also be incorporated as the anomaly scoring framework and SHAP can add the explainability element that the paper lacked.

Third, the inherent limitations outlined in the base study directly mirror the specified gaps in the study areas. The base authors indicate that execution time was an important parameter, as such a method would encourage the study of a more efficient algorithm that is better suited to a faster and more accurate implementation, and is capable of processing the sheer amount of data stream in real-time conditions in large organisations. The base study did not carry out any validation in accordance with an publicly benchmarkable public data, so the use of the Kaggle Cold Chain Shipment Silent Failure Data Set ensures comparability and reproducibility of the project results. It may as a result, help others to identify anomalies and potentially cause disruptions within other projects involving such data. This combination directly addresses Gaps 1, 2, 3 and 5 which have been identified within this literature review.

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